

Month 1: Requirements Analysis, Environment Setup, and Dataset Exploration

This month focused on feasibility studies, establishing the development environment, acquiring target datasets, and defining system requirements.

Week 1: Feasibility Study and Repository Setup

- **Progress Summary:** Technical, economical, and operational feasibility studies were conducted. A Python 3.10 virtual environment was established, and a Git repository was initialized with standard folder structures (src/, data/, reports/). A .gitignore file was configured to protect local environment variables and credentials.

Week 2: Dataset Acquisition and Structure Analysis

- **Progress Summary:** The HotpotQA benchmark dataset was acquired and imported. Python parsing scripts were developed to explore the JSON schema of multi-hop questions, target answers, and Wikipedia document passages, mapping logical relationships between entities.

Week 3: System Objectives and Requirement Definition

- **Progress Summary:** The system objectives and Software Requirement Specifications (SRS) were defined. Functional, performance, dependability, and security requirements were documented, establishing the target metrics for retrieval recall, query latency, and database persistence.

Week 4: Code Architecture and Standards Setup

- **Progress Summary:** The coding standards and modular architecture design were established. Baseline imports, dependency trees, and package configurations were set up, preparing modular files for data ingestion, models, database adapters, and visualization.